Matteo Nurisso, Ph.D.

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Employment

2024 – present **Assegno di ricerca Post Dottorale (Post-Doc)** Consiglio Nazionale delle Ricerche, Institute of Atmospheric Sciences and Climate (CNR-ISAC), Turin.
Topic: EPOCHAL project about paleoclimate simulations and code development of AQUA in the Destination Earth project.

2022 – 2024 **Assegno di ricerca professionalizzante** Consiglio Nazionale delle Ricerche, Institute of Atmospheric Sciences and Climate (CNR-ISAC), Turin.
Topic: Teleconnections diagnostic and code development of the framework for the AQUA code in the Destination Earth project.

Education

2018 – 2022 **Ph.D., Scuola Internazionale di Studi Superiori Avanzati (SISSA)** Astrophysics and Cosmology.
Thesis title: *Particles acceleration with Magnetic Reconnection in large scale MHD simulations*.
Supervisors: *Annalisa Celotti and Andrea Mignone*

2014 – 2018 **Master Degree , University of Turin** Astrophysics and Theoretical Physics.
Thesis title: *Anisotropic Cosmic Ray transport in the galaxy with PLUTO*.
Supervisors: *Andrea Mignone and Nicolao Fornengo*

2011 – 2014 **Bachelor Degree , University of Turin** Physics.
Thesis title: *The Heisenberg model and the Bethe Ansatz*.
Supervisor: *Marco Billó*

Research Publications

1. Suriano, A., **Nurisso, M.**, Celotti, A., Mignone, A., & Bodo, G. (In preparation). Particles acceleration with magnetic reconnection in large scale mhd simulations: Ii. spectra update.
2. John, A., Beyer, S., Athanase, M., Benítez, A. S., Goessling, H., Hossain, A., … Jung, T. (2024). Global storyline simulations at the kilometre-scale. doi:10.22541/essoar.173160166.64258929/v1
3. Caprioli, S., von Hardenberg, J., Ghinassi, P., Ghosh, S., Kluft, L., Koldunov, N., … Davini, P. (2024). Aqua: A novel quality assessment tool for km-scale simulations in the destination earth climate digital twin - the diagnostics suite. doi:10.5194/egusphere-egu24-11230
4. Nazarova, N., von Hardenberg, J., Davini, P., **Nurisso, M.**, Caprioli, S., & Ghinassi, P. (2024). Evaluating tropical precipitation extremes: Insights from first simulations from nextgems and destination earth climate digital twin. doi:10.5194/egusphere-egu24-11498
5. **Nurisso, M.**, von Hardenberg, J., Caprioli, S., Ghosh, S., Koldunov, N., P. Kinoshita, B., … Davini, P. (2024). Aqua: A novel quality assessment tool for km-scale simulations in the destination earth climate digital twin - the core framework. doi:10.5194/egusphere-egu24-10935
6. **Nurisso, M.**, Celotti, A., Mignone, A., & Bodo, G. (2023). Particles acceleration with magnetic reconnection in large scale RMHD simulations: I. current sheet identification and characterization. *Monthly Notices of the Royal Astronomical Society*. doi:10.1093/mnras/stad1348
7. **Nurisso, M.**, Celotti, A., Mignone, A., & Bodo, G. (2022). Particle acceleration via magnetic reconnection in large-scale MHD jet simulations. (Vol. 17, pp. 54–55). doi:10.1017/s1743921323000935

8 **Nurisso, M.** (2022). *Particles acceleration with magnetic reconnection in large scale mhd simulations* (Ph.D thesis). Retrieved from https://hdl.handle.net/20.500.11767/130671

9 **Nurisso, M.** (2018). Anisotropic cosmic ray transport in the galaxy with pluto. Retrieved from https://api.semanticscholar.org/CorpusID:221133901

Projects

2025 – present
- PI of the Special Project Earth system modeling of PaleOClimatic HyperthermALs (EPOCHAL) at ECMWF HPC

2024 – present
- Participation as Post-Doc in the PRIN: **Earth system modeling of PaleOClimatic HyperthermALs** (EPOCHAL) founded by the NextGeneration EU.
- Participation in the WP14 and co-lead of Activity 8 (A8 corresponding to WP7 of the Phase 1) of the **Destination Earth Climate Digital Twin** project Phase 2, a flagship initiative of the European Commission.

2022 – 2024
- Participation in the WP7 of the **Destination Earth Climate Digital Twin** project Phase 1, a flagship initiative of the European Commission.

Code contributions

2022 – present
- **AQUA (Application for QUality Assessment)**. Developer of the code framework and the teleconnections and timeseries diagnostics.

2024 – present
- **Stece code**. A collection of script and tools for EC-EARTH4.

2017 – 2023
- **PLUTO code**. Developer of a module for magnetic reconnection particles acceleration in Magneto-Hydrodynamical simulations. The module is now integrated in the gPLUTO implementation of PLUTO code for GPUs.

Conferences and Hackatons

2024
- **Destination Earth Climate DT Project meeting and Hackathon** (Hamburg, Germany)
- **nextGEMS Cycle 4 Hazard hackathon** (Wageningen, Netherland)
- **Destination Earth Climate DT Final Kick-off** (Helsinki, Finland)
- **EGU General Assembly 2024** (Vienna, Austria) (Abstract: https://meetingorganizer.copernicus.org/EGU24/EGU24-10935.html)
- **EC-Earth General Assembly** (De Bilt, Netherlands)

2023
- **2nd Destination Earth User eXchange** (Bonn, Germany) (Poster from AQUA code: http://bit.ly/aqua-poster)
- **nextGEMS Cycle 3 Hackathon** (Madrid, Spain)
- **Destination Earth Hackathon** @ BSC (Barcelona, Spain)

2022
- **7th Heidelberg International Symposium on High-Energy Gamma-Ray Astronomy** (Barcelona, Spain)
- **Active Galactic Nuclei XIV: The Renaissance of Black Holes and Galaxies** (Florence, Italy)

2020
- **EuroScience Open Forum 2020** (Trieste, Italy)

2019
- **International Conference of Physics Students 2019** (Cologne, Germany)

2017
- **International Conference of Physics Students 2017** (Turin, Italy)

Skills

Languages	Strong reading, writing and speaking competencies for English and Italian. Moderate reading, writing and speaking competencies for French.
Coding	C, C++, Python (xarray, dask), Fortran
Software	Git, Github, Linux, Slurm, Docker, Singularity, EC-Earth ESM code, CDO, CI/CD, MPI, OpenMP, Cuda, IDL, Visit, Html, PLUTO MHD code
Scientific	Climate simulations, Magnetohydrodynamic simulations, Academic research, LaTeX typesetting and publishing.

Miscellaneous Experience

Seminars

2024
- AQUA tool introduction - A workshop on AQUA usage @ **nextGEMS cycle 4 Hazard hackathon**
- AQUA: a novel quality assessment tool for km-scale simulations in the Destination Earth Climate Digital Twin – An introduction to AQUA and to nextGEMS data access @ **nextGEMS Storms, Eddies and Science hour**
- AQUA - a novel quality assessment tool for km-scale simulations in the Destination Earth Climate Digital Twin – The Core Framework @ **EGU General Assembly 2024**

Teaching

2017 – 2018
- **Tutor** of "Introduzione alla programmazione", University of Turin

Workshops

2023
- **CodeRefinery workshop**: Introduction to Git, Collaborative distributed version control, Reproducible and tested code and research

2022
- **SCtrain training week**: Introduction to Parallel Programming

2021
- **Machine Learning Lectures** Cluster of Galaxies IFPU group

2020
- **Deep Learning Day** AI Student Society

Science Communication

2022
- **Social media management** and **Personal Branding**
- **Science Communication:** the use of objects and easy experiments in face-to-face events

2021
- **Science Communication** mini-course

2019
- **Training and Sharing**: Best practices in Science Communication
- **Science Dialogues**: training course on Science Communication

Outreach

2020-2022
- **Trieste Next** volunteer

2018 – 2022
- **Sissa4Schools** volunteer

2018-2021
- **SISSA Student Day** speaker

2017
- **ICPS** volunteer (International Conference of Physics Students)

Miscellaneous Experience (continued)

2013 – present

- Vicepresident (2014-2023) and President (since 2024) of **Silviadizenzero**, active in Climate Science outreach